

Mathematics

A Cultural Heritage

Lecture 4 (Part 2)

Funny Side of Mathematics

Course Topics

*Funny Side of
Mathematics*

Mathematics,
literatures, movies,
and more

*Mathematics
and the Arts*

Mathematics
and Education

Two mathematics
cultures and a few
mathematical lives

Numbers

Numbers
at work

Our Obsessions
with Numbers

**Mathematics
and Society**

Mathematics and Truth
(Philosophical Aspects
of Mathematics)

**THEOREMS (PROBLEMS)
AND PROOFS**

The Funny Side of Mathematics

- People usually perceive mathematics as a serious, no-nonsense subject. In fact a recent research shows that mathematics can really give some people headaches!

UChicagoNews THE UNIVERSITY OF CHICAGO

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When people worry about math, the brain feels the pain

By William Harms
OCTOBER 31, 2014

Mathematics anxiety can prompt a response in the brain similar to when a person experiences physical pain, according to new research at the University of Chicago.

Using brain scans, scholars determined that the brain areas active when highly math-anxious people prepare to do math overlap with the same brain areas that register the threat of bodily harm—and in some cases, physical pain.

"For someone who has math anxiety, the anticipation of doing math prompts a similar brain reaction as when they experience pain—say, burning one's hand on a hot stove," said [Sian Beilock](#), professor of psychology at the University of Chicago and a leading expert on math anxiety.

Surprisingly, the researchers found it was the anticipation of having to do math, and not actually doing math itself, that looked like pain in the brain. "The brain activation does not happen during math performance, suggesting that it is not the math itself that hurts; rather the anticipation of math is painful," added Ian Lyons, a 2012 PhD graduate in psychology from UChicago and a postdoctoral scholar at Western University in Ontario, Canada.

The two report their findings in a paper, "When Math Hurts: Math Anxiety Predicts Pain Network Activation in Anticipation of Doing Math," in the current issue of *PLoS One*.

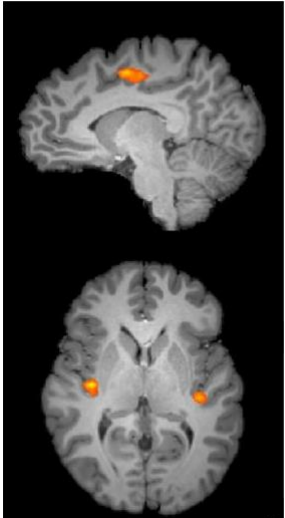
For the study, the scholars worked with 14 adults who were shown to have math anxiety based on their responses to a series of questions about math. The questions gauged one's anxiety when receiving a math textbook, walking to math class or realizing math requirements for graduation. Additional tests showed that these individuals were not overly anxious in general; instead, their heightened sense of anxiety was specific to math-related situations.

The study volunteers were tested in an fMRI machine, which allowed researchers to examine brain activity as they did math. *Volunteers experienced mathematics-related anxiety, for example, the*

4961 526 11.1K

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PHOTOS



Sian Beilock

The Funny Side of Mathematics

The following are some
jokes, cartoons, videos, etc.,
to show that mathematics can
sometimes be

Fun!

Math is Precise

A joke on mathematicians:

An **astrologer**, a **chemist**, and a **mathematician** on a bus during their first visit to Scotland. They see a **black** sheep grazing alone in a pasture as they drive by.



The **astrologer** excitedly exclaims, “Ah, this shows Scottish sheep are black!”

The **chemist** didactically corrects him: “No, no, it just shows some Scottish sheep are black.”

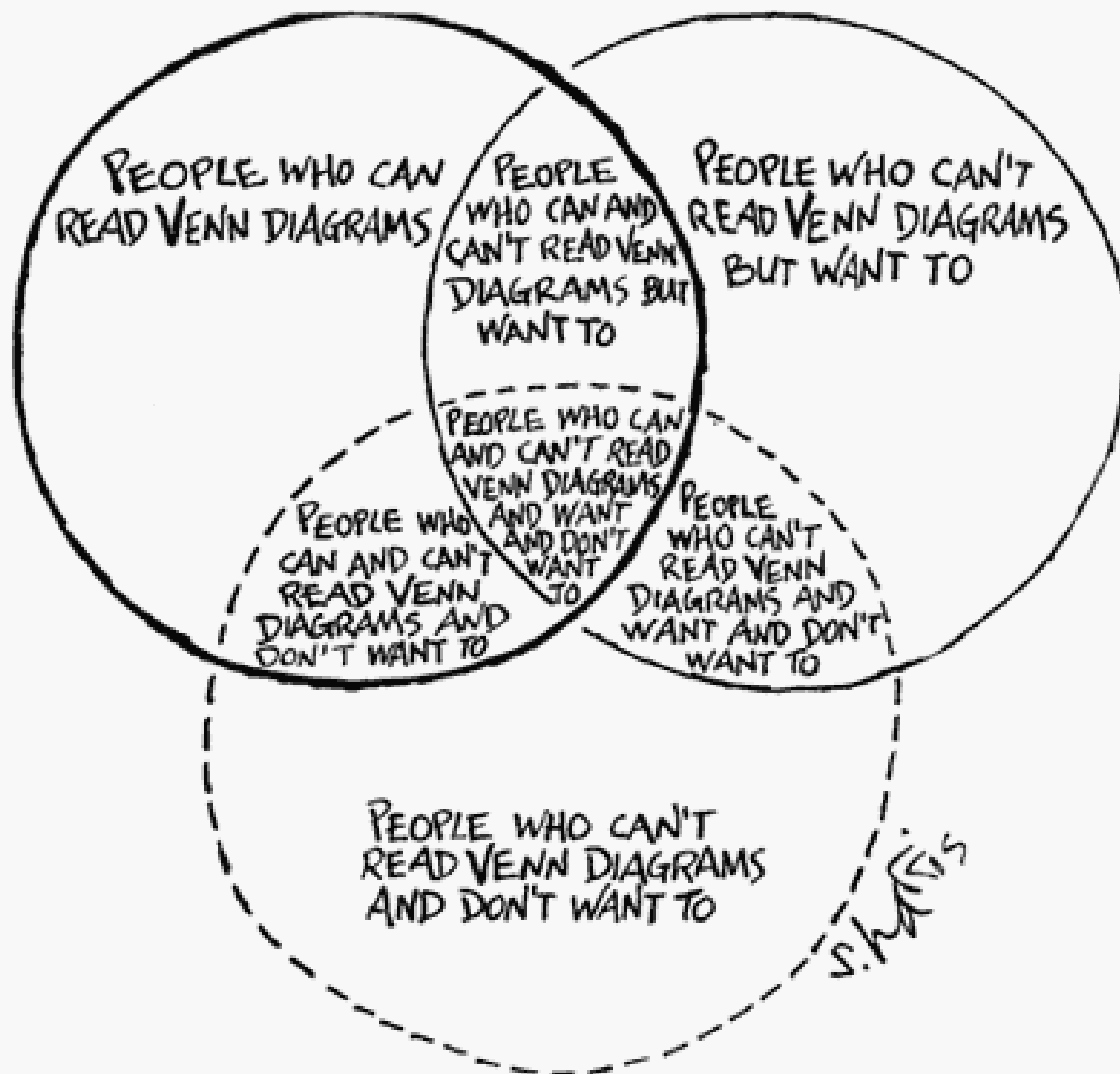
Math is Precise



The **mathematician** then says, “Actually, we can only be sure **there is at least one Scottish sheep of which at least one side is black.**”

Two "Math for Dummies" at
\$16.99 each. That'll be \$50.

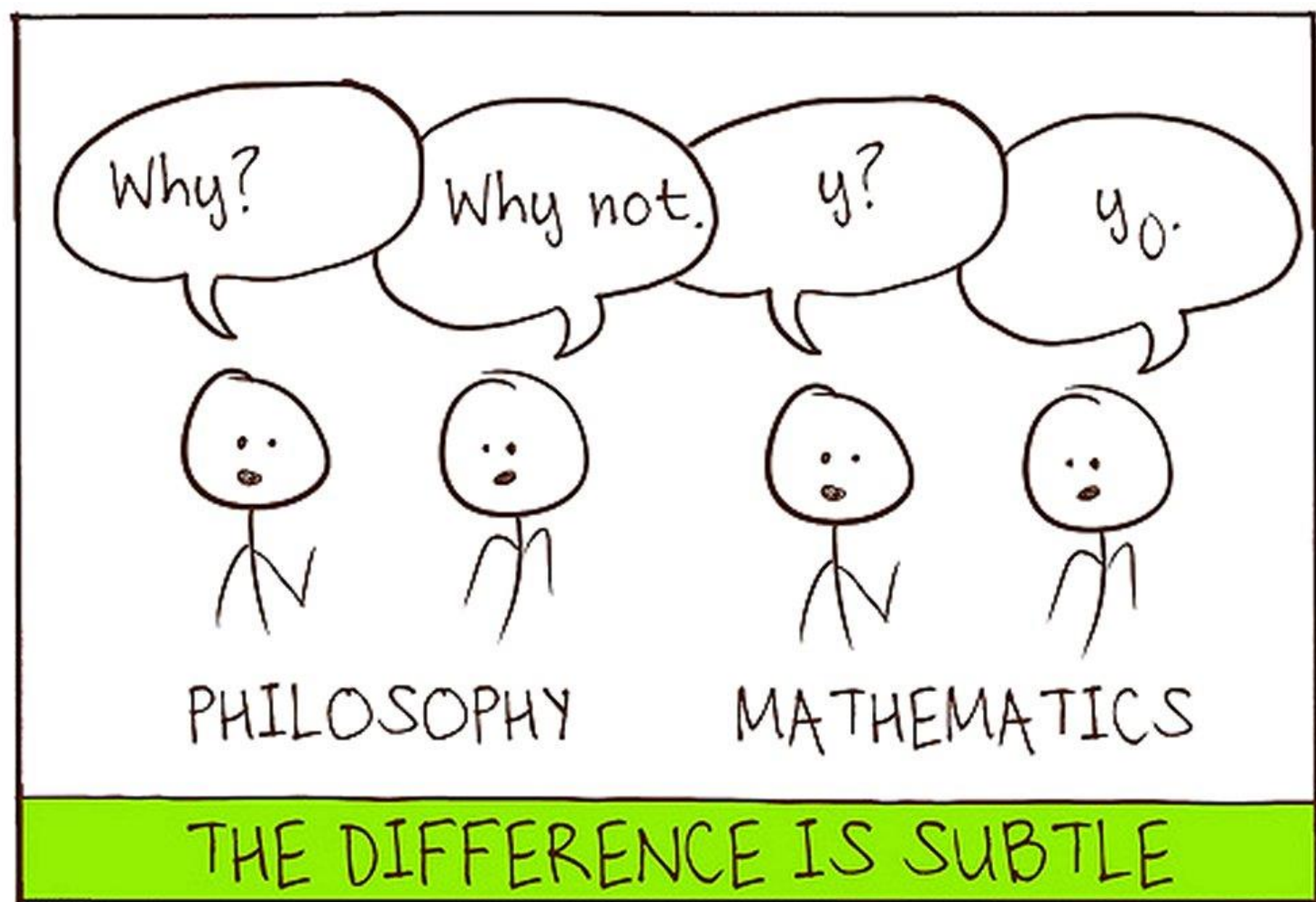




From designers
to all clients.







- Weather is cold these days.
- Here is a helpful advice from a father.



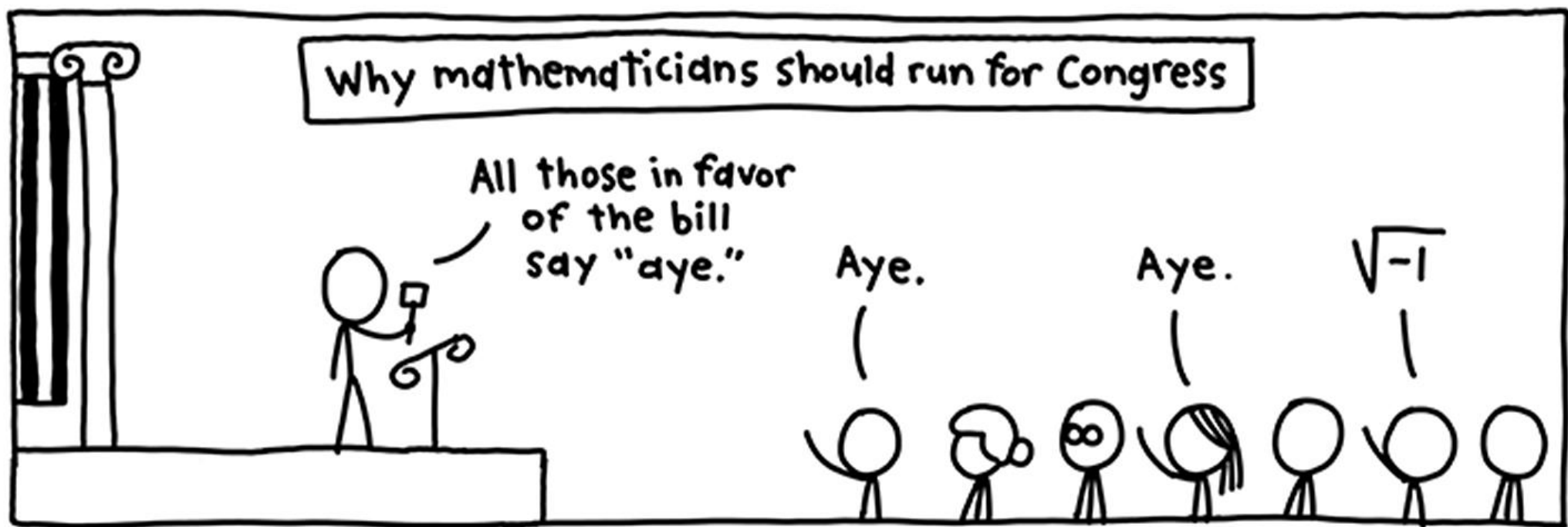
Why mathematicians should run for Congress

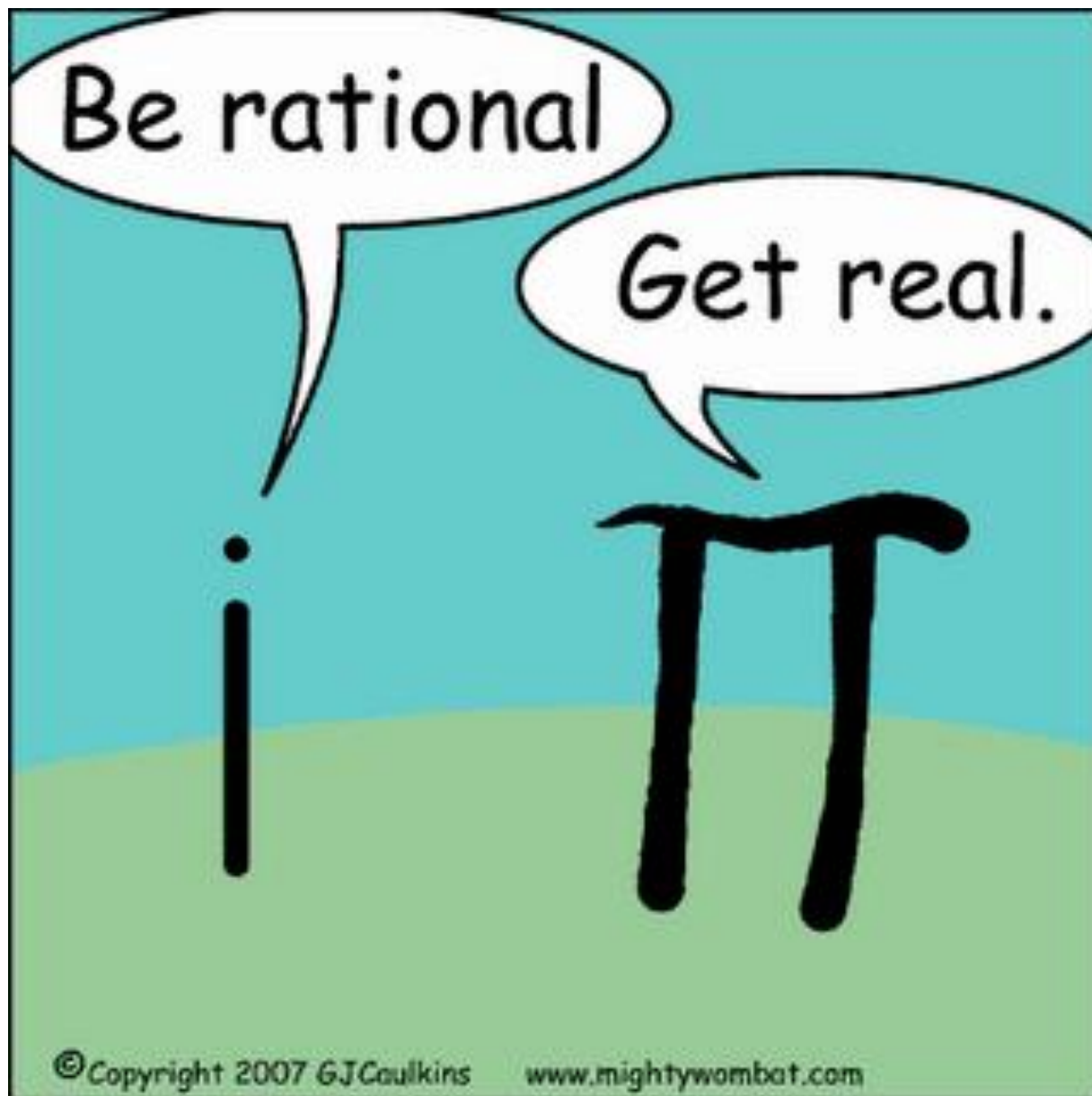
All those in favor
of the bill
say "aye."

Aye.

Aye.

$\sqrt{-1}$





數學 MATH

Mental Abuse To Humans

人類精神虐待

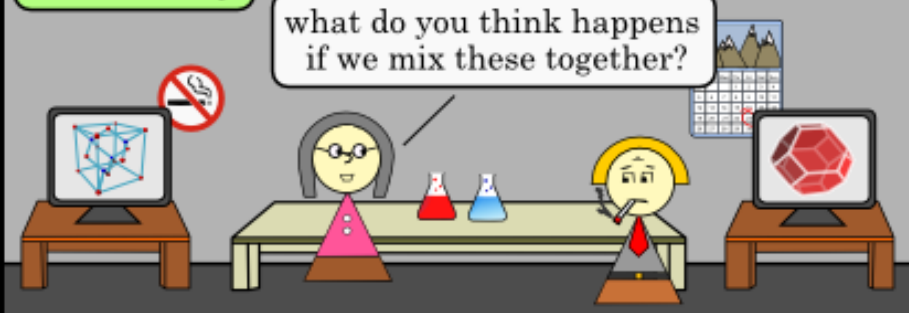
Biology



Physics



Chemistry



Mathematics



spikedmath.com
© 2010



meanwhile... the rest of the world goes on as usual

*“There are **10** kinds of people in the world — those who understand binary and those who don't.”*

*“There are **10** kinds of people in the world — those who understand trinary, those who don't understand trinary and those who mistake it for binary.”*

	1
	0

BINARY
SU DOKU



**Alcohol and calculus
don't mix.
*Never drink and derive.***

©2009 Internet Alcoholics

Hi, Dr. Elizabeth?
Yeah, uh... I accidentally took
the Fourier transform of my cat...



After explaining to a student through various lessons and examples that:

$$\lim_{x \rightarrow 8^+} \frac{1}{x - 8} = \infty$$

I tried to check if the student really understood, so I gave her another example.

This was the result:

$$\lim_{x \rightarrow 5^+} \frac{1}{x - 5} = 5$$

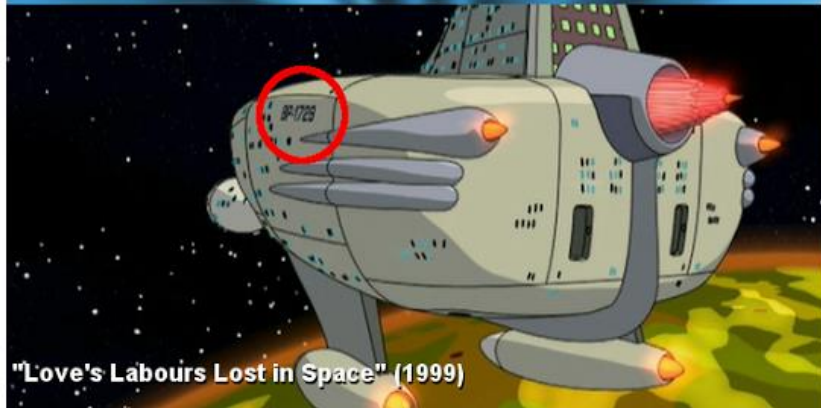


Screenshot: Futurama, Comedy Central

FUTURAMA

1729 IS A MAGIC NUMBER

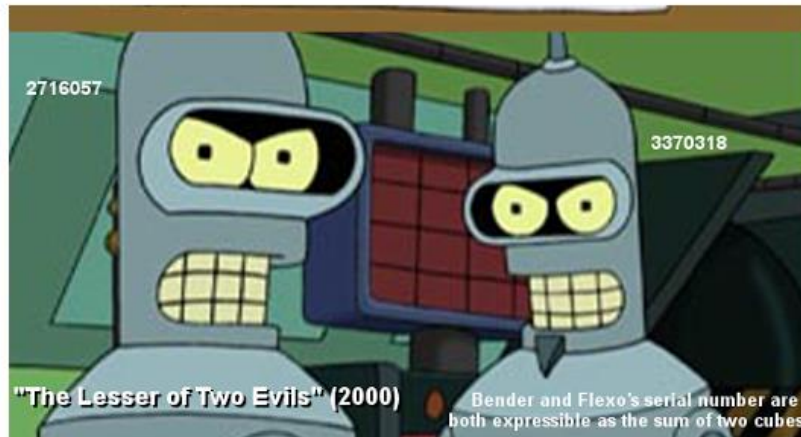
The number 1729 is known as the "Hardy-Ramanujan number" based on a story told by G. H. Hardy when he visited Srinivasa Ramanujan in the hospital. It is the smallest number expressible as the sum of two cubes in two different ways.



"Love's Labours Lost in Space" (1999)



"Xmas Story" (1999)



"The Lesser of Two Evils" (2000)

Bender and Flexo's serial number are both expressible as the sum of two cubes



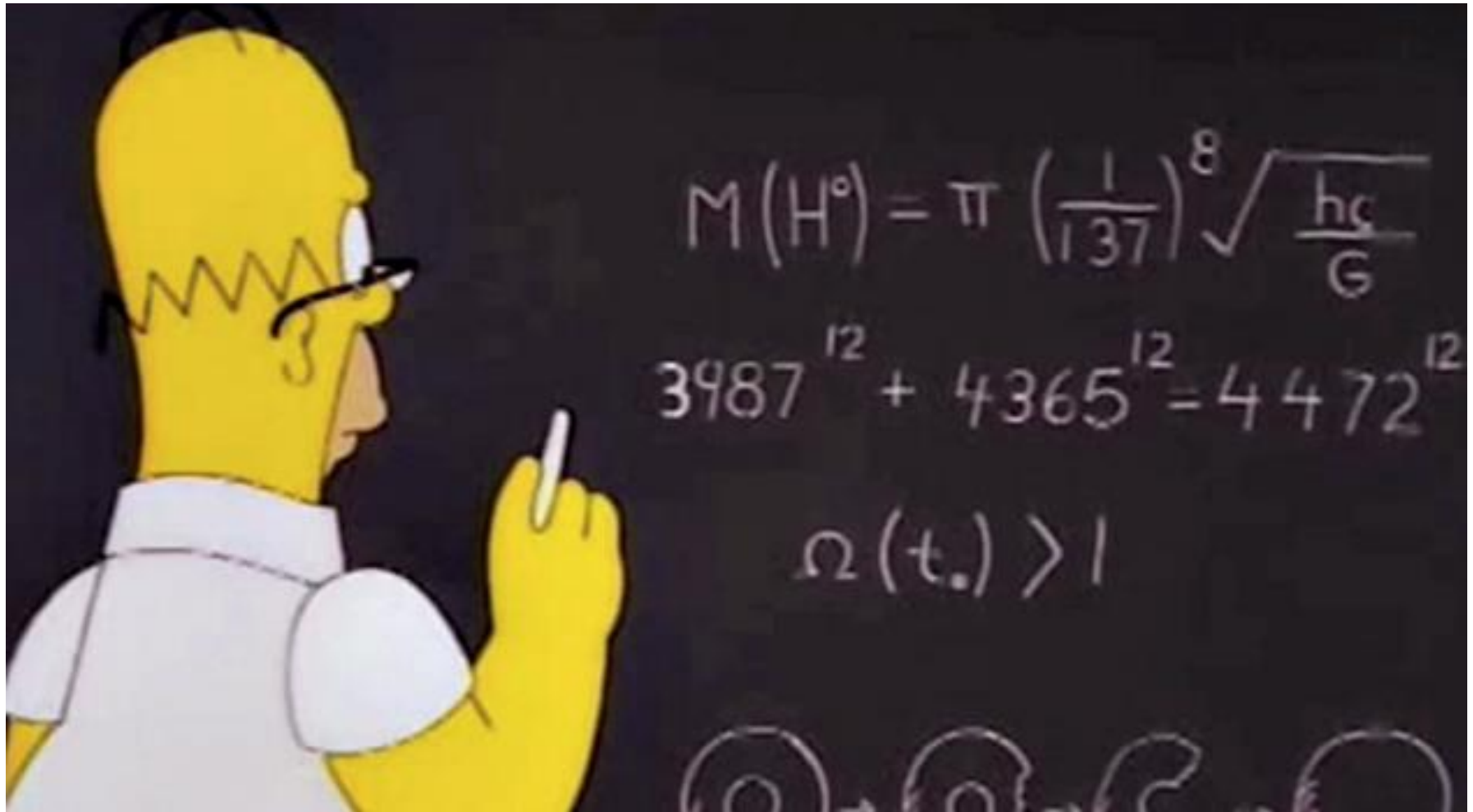
"The Farnsworth Parabox" (2003)



"Bender's Big Score" (2007)

87539319 is the smallest number expressible as the sum of two cubes in three different ways

www.thegeektwins.com



Homer's insights into math and physics

Screenshot: [Huffington Post](#)/Fox

Video

- What is 7 times 13?



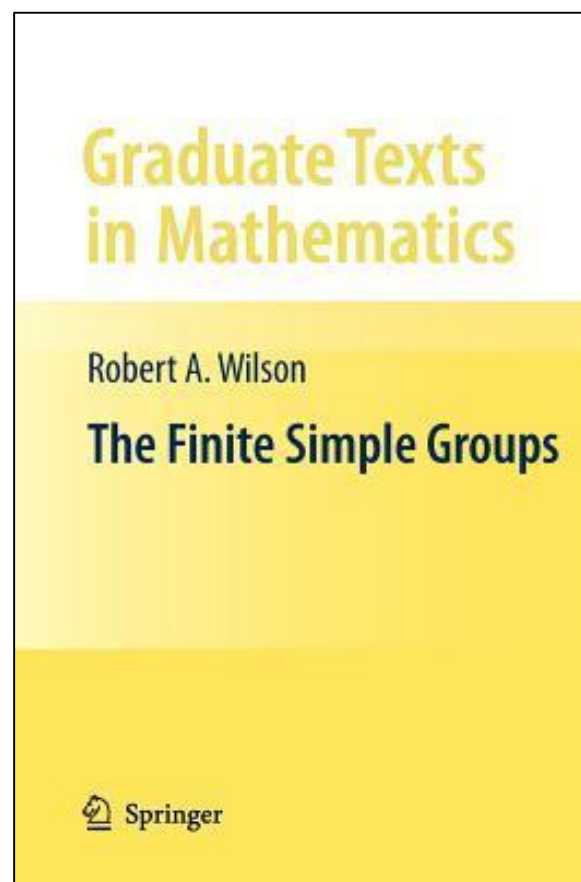
Video

- A mathematical love song. 



Finite Simple Groups

There is actually
such a term “**Finite
Simple Groups**” in
mathematics!



Mathematical terms in the song:

Associative · Axiom of Choice · Bundle · Chains · Class ·
Complexified · Continuous · Corollary · Dense · Directed ·
Domain · Equivalence · Finite group · Finite limit ·
Forgetful functor · Free group · Identity · Image · Kernel ·
Map into · Mirror pair · One-to-one · Open · Operator ·
Order of group · Path · Proposition · Quotient · Rank-one map ·
Relation · Separable · Simple group · Simply connected ·
Smooth · Stable · Tensor · Two-form · Upper bound · Wedge ·
Well-defined · Without loss of generality · Zero

Finite Simple Group of Order Two

http://www.youtube.com/watch?v=UTby_e4-Rhg

You may want to read the lyrics.

Conversation between a Couple

Before marriage...

1. He: Yes. At last. It was so hard to wait.
2. She: Do you want me to leave?
3. He: No! Don't even think about it.
4. She: Do you love me?
5. He: Of course! Always have and always will!
6. She: Have you ever cheated on me?
7. He: No! Why are you even asking?
8. She: Will you kiss me?
9. He: Every chance I get.
10. She: Will you hit me?
11. He: Hell no! Are you crazy?
12. She: Can I trust you?
13. He: Yes.
14. She: Darling!

After marriage... read from bottom to top.

How Does it Work?

- Some analysis:

Question

Answer

Question

Answer

Question

Answer

Question

Answer

Question

Answer

Question

Answer

Reading Assignments

The following will be posted on the Moodle site.

1. "A Defense of Quadratic Equations," UK House of Commons Daily Debates, 26 June 2003
2. "Is Algebra Necessary" by Andrew Hacker, in New York Times, July 28, 2012 (and 477 comments by readers)